2.12 Rank of a Matrix

- The rank of the matrix refers to the number of linearly independent rows or columns in the matrix. ρ(A) is used to denote the rank of matrix A.
- A matrix is said to be of rank zero when all of its elements become zero.
- The rank of the matrix is the dimension of the vector space obtained by its columns.
- The rank of a matrix cannot exceed more than the number of its rows or columns. The rank of the null matrix is zero.
- The nullity of a matrix is defined as the number of vectors present in the null space of a given matrix. In other words, it can be defined as the dimension of the null space of matrix A called the nullity of A. Rank + Nullity is the number of all columns in matrix A.

  A real Number 'r' is said to be the rank of a matrix '$A_{m\times n}$' if

  (1) There is at least one square sub-matrix of A of order *r* whose determinant is not equal to zero.

  (2) If the matrix A contains any square sub-matrix of order (*r* + 1) and above, then the determinant of such a matrix should be zero.

  It is mathematically denoted by $\rho(A) = r$

2.12.1 Properties of Rank of a Matrix

(i) $\rho(A_{m\times n}) \leq (m, n)$

(ii) $\rho(AB) \leq min\{\rho(A), \rho(B)\}$

(iii) Rank of transpose of matrix is equal to rank of matrix

(iv) Elementary operations do-not affect the rank the matrix

(v) $\rho(A + B) \leq \quad \{\rho(A) + \rho(B)\}$

2.12.2 Row Echelon Form

A Matrix $A_{m\times n}$ is said to be in row-echelon form if

(i) Number of zeroes before the 1st Non-zero element in any row is less than the number of such zeroes in its succeeding row.

(ii) Zero rows (if any) should lie at the bottom of the Matrix.

$\rho(A_{m\times n})$ = Number of non-zero rows in the Row-Echelon form of A.

2.13 System of Equations

The given system of equations

$$a_{11}x_1 + a_{12}x_{12} + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

can be written in Matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}} \boxed{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \boxed{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}$$

Ax = B

Coefficient Matrix — Variable Matrix — Constants Matrix

The system Ax = B is said to be a homogeneous system if B = 0.

The system of Ax = B is said to be a non-homogeneous system if $B \neq 0$.

2.13.1 Consistency of a non-homogeneous system of Equations

For the above system of non – homogeneous equations, Ax = B; Augmented Matrix = [A/B] = $\begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$

(i) If $\rho(A) = \rho(A/B) =$ Number of unknowns, then the system Ax = B has a unique solution.

(ii) If $\rho(A) = \rho(A/B) <$ Number of unknowns, then the system has infinitely many solutions.

(iii) If $\rho(A) \neq \rho(A/B)$, then the system has no solution.

Number of linearly independent solutions for a system of 'n' equations given by Ax = B is $n - \rho(A)$

2.13.2 Consistency of Homogeneous System of Equations

$a_{11}x + a_{12}y + a_{13}z = 0$

$a_{21}x + a_{22}y + a_{23}z = 0$

$a_{31}x + a_{32}y + a_{33}z = 0$

$$\text{Ax = 0} \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \; [A/B] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \end{bmatrix}_{3\times 4}$$

If $\rho(A) = \rho(A/B) = n\ (i.e\ |A| \neq 0)$; the system has a unique solution.

(Trivial solution; x = 0, y = 0, z = 0)

If $\rho(A) = \rho(A/B) < n(|A| = 0)$; the system has infinitely many solutions (Non-trivial solution exists for the system).

2.14 Linear Combination of Vectors

If $x_1, x_2, x_3, \ldots\ldots, x_n$ are 'n' rows vectors, then the combination $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n$ is called a linear combination of $x_1, x_2, \ldots., x_n (k_1, k_2, k_3, \ldots\ldots k_n$ are scalars)

(1) The linear combination $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n$ is said to be linearly dependent if $k_1x_1 + k_2x_2 + k_3x_3 + \ldots\ldots + k_nx_n = 0$ when $k_1, k_2, k_3, \ldots\ldots, k_n$ (NOT All zeroes).